Transfer task: plan and begin

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Organize the given evidence and begin an independent position prediction.

SUCCESS CRITERIA - I can...

1. Organize the task data with fixed axes and matching units.
2. Calculate central and bounded final coordinates.
3. Save an explicit plan for the graph and explanation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A 3 kg lamp hangs at rest from a vertical rope. Use g=9.8. Find tension. Then find tension if it accelerates upward at 2 m/s².

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $v_x = v \cos \theta$; $v_y = v \sin \theta$; $x_f = x_i + v_x t$ $y_f = y_i + v_y t$.
Fictional fixed-Earth model: $x_i = -1.2 \times 10^8$ km; $y_i = 0$; $v = 8500$ m/s; $\theta = 15°$ from +x toward +y; $t = 1.6 \times 10^7$ s. Convert units.

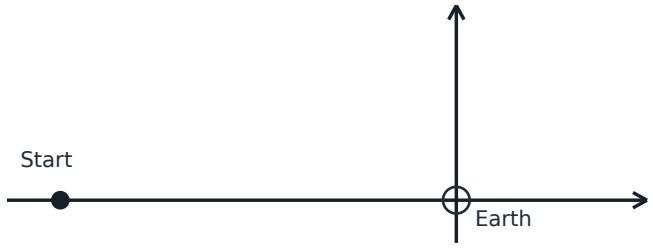
Worked reasoning: A toy model uses x_i=0 m and constant v=2 m/s for 5 s, giving x=10 m. This is a conditional prediction, not proof that the real object follows the model.

Your turn: Draw the predicted path and mark one observation you would collect to test the constant-velocity assumption.

Marzano check - circle: ① ② ③

03 Plan and represent

Draw the coordinate convention and write an ordered plan for units, velocity components, endpoint, bounds, graph and public explanation.



The small starting dot is the object; the labeled circle marks Earth at the origin. Not to scale.

Marzano check - circle: ① ② ③

04 Central prediction

Resolve velocity and calculate final x, y in km. Distinguish final position from displacement and separation. Keep unrounded components until the end.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Propagate bounded inputs

Initial x bound±50, 000 km; speed bound±200 m/s at fixed 15°. Initial x and speed vary independently; angle, y_i, time exact. Find coordinate half-widths and intervals.

Marzano check - circle: 1 2 3

06 Choose a method

A puck follows a circle at a steady 7 m/s. A student calls it equilibrium. Draw velocity at two points and correct the claim.

Marzano check - circle: 1 2 3

07 Checkpoint

Prepare central/lower/upper x(t) values at t 0 and final time for tomorrow’s graph. Write one assumption, one excluded physical effect, and the next unfinished step.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A model predicts x=10 m while the measurement is 8.0±0.2 m. Construct one constant-velocity model and one uniformly accelerated model from rest that fit a 5 s trip to 8 m. Explain what more data would resolve.

Marzano check - circle: 1 2 3